

Investigation of the potential use of Virtual Reality for Agoraphobia exposure therapy

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Figure 1: Left: Virtual Reality Environment: Park. Right: Virtual Reality Environment: Interacting with neighbour

ABSTRACT

Preliminary research study to evaluate the potential and need for virtual reality in the mental health sector specifically focusing on the treatment of agoraphobia. A survey was sent to numerous participants that have been diagnosed and currently receiving treatment for agoraphobia. Results have concluded there is a demand for virtual reality treatment for agoraphobia and this in turn can lead to future studies into the VR therapy.

Index Terms: Human-centered computing—Accessibility—Accessibility systems and tools; Human-centered computing—Human computer interaction—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction—Interaction paradigms—Collaborative interaction;

1 INTRODUCTION

The purpose of this preliminary research study is to assess the potential benefits of the use of virtual reality as an aid to assist the treatment for patients diagnosed with agoraphobia [1], the fear and avoiding of open spaces and situations that can make the person feel trapped and cause anxiety. Past research papers have shown the benefits of virtual reality used in the medical field as a form of treatment aid for patients in numerous areas. [3] [6] [5] A survey was conducted and sent to individuals who are currently undergoing treatment for agoraphobia. The responses show that participants feel virtual reality would be a beneficial aid with treatment. Virtual reality treatments aids for agoraphobia can be potentially beneficial for patients who have severe anxiety, it could help expose the patient to different environments while being in a safe space and assist the patient in learning coping mechanisms for anxiety and panic attacks. The virtual reality environment will have several scenarios: inside the home, driving, public transport, open spaces i.e. parks, crowded areas such as supermarkets. Each scenario can cater to a patient's different needs and requirements for their treatment plan. These scenarios can potentially help the patient to gradually expose them to stressful situations in a safe environment and help them tackle what triggers their anxiety while learning ways to help ease their anxiety.

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2 MOTIVATION

Agoraphobia is a type of anxiety disorder in which you fear and/or avoid places or situations that can cause panic. It could make you feel trapped and helpless. Ireland has one of the highest rates of mental health illnesses within Europe. Approximately 18.5% of the population in Ireland have recorded having a mental health disorder such as anxiety, depression, alcohol or drug use, etc. [4] [2] This project is a virtual reality simulation that is aimed to help those suffering from agoraphobia by gently exposing them to everyday scenarios, such as traveling by public transport, grocery shopping and walking through a park.

3 STAKEHOLDERS

This study will help individuals suffering from agoraphobia. It will be used to help patients use forms of CBT (cognitive-behavioural therapy) and exposure therapy as opposed to traditional forms of therapy with a medical professional. Medical professionals, such as therapists, doctors, and psychiatrists can use it as a form of treatment to help patients. [7] The medical professionals can follow a handbook provided to customise the treatment plan for each patient.

4 SYSTEM DESIGN

In order to gather the required information a system was designed to gather the required information. The following outlines the experimental design used for this study and to gather the data needed to determine the communal need for this type of application.

The above diagram outlines the experimental apparatus used. Future studies will expand upon this design and allow for greater exposure elements with additional haptic devices and sensors.

5 GAMEPLAY

The patient can use the sensor controllers to move and interact within the environment around them. The patient will be given tasks to complete in each level. As you progress, the tasks will gradually become more difficult to complete and the player will be pushed further and further away from their comfort zone. This steady increase in the level of difficulty will gradually allow the user to ease into the exposure therapy gradually and at their own pace. This allows for better user comfort and usability of the application.

6 PRELIMINARY SURVEY RESULTS

A survey was conducted online and sent to participants that have been diagnosed with agoraphobia. This survey asked the participants questions relating to what they would like to see in the scenarios, potential areas of focus, what areas personally affect them. The survey has shown that 76.47% of participants have some anxiety

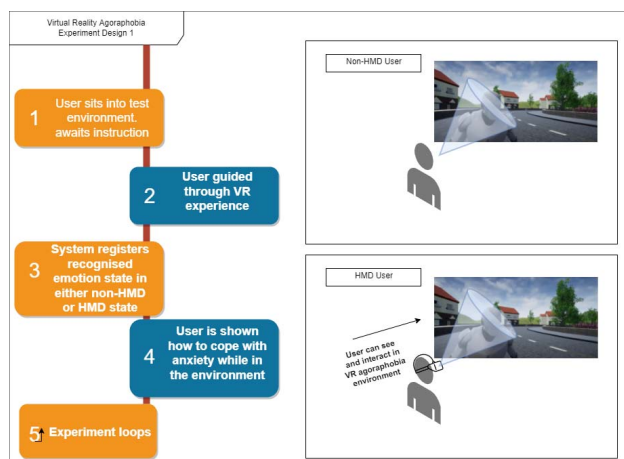


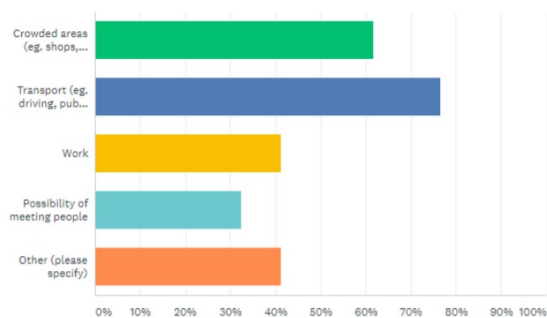
Figure 2: System Design

relating to transport (Figure 3). A further survey can be conducted to analyse how many participants are affected by public and private transport. 61.76% of participants have anxiety relating to crowded areas, i.e. supermarkets (Figure 3). This can be another key area to focus on and centralise a unique scenario around this area. The survey shows that 41.18% of participants have anxiety relating to work (Figure 3). A further study can be conducted to look further into this, i.e. what specifically is causing the anxiety, potentially colleagues, workload or other work related stresses.

Q3

Which areas affect you most?

Answered: 34 Skipped: 0



ANSWER CHOICES	RESPONSES	
Crowded areas (eg. shops, lines, concerts)	61.76%	21
Transport (eg. driving, public transport, taxi)	76.47%	26
Work	41.18%	14
Possibility of meeting people	32.35%	11
Other (please specify)	41.18%	14
Total Respondents: 34		

Figure 3: Survey Results

7 FUTURE WORK

The project can be developed even further by creating more unique scenarios for the patients to interact with. These different scenarios can focus on public transport, vast environments such as parks,

forests and mountain terrains, social situations such as grocery shopping, meetings. User testing can be conducted to refine the project. Ultimately it can be produced for the medical industry to use alongside other medical treatments. Due to the COVID-19 global pandemic on premises access to these medical professionals has been severely limited. Thus the growing need for tele-medicine and remote/home based therapies to be administered by the medical expert.

8 CONCLUSION

Past research studies has shown the benefits of using virtual reality as an aid for treatment. Virtual reality can be a very beneficial aid to assist in the treatment of agoraphobia. The survey results show that numerous areas can be focused on to aid the treatment of agoraphobia within virtual reality. These key areas of focus are the following: open spaces i.e. parks, transport, public and private, and work environments. Further research would need to be conducted to help create more detailed scenarios focusing on these key issues. We would seek to develop further on these concepts in future works and present our results in this promising area of research.

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